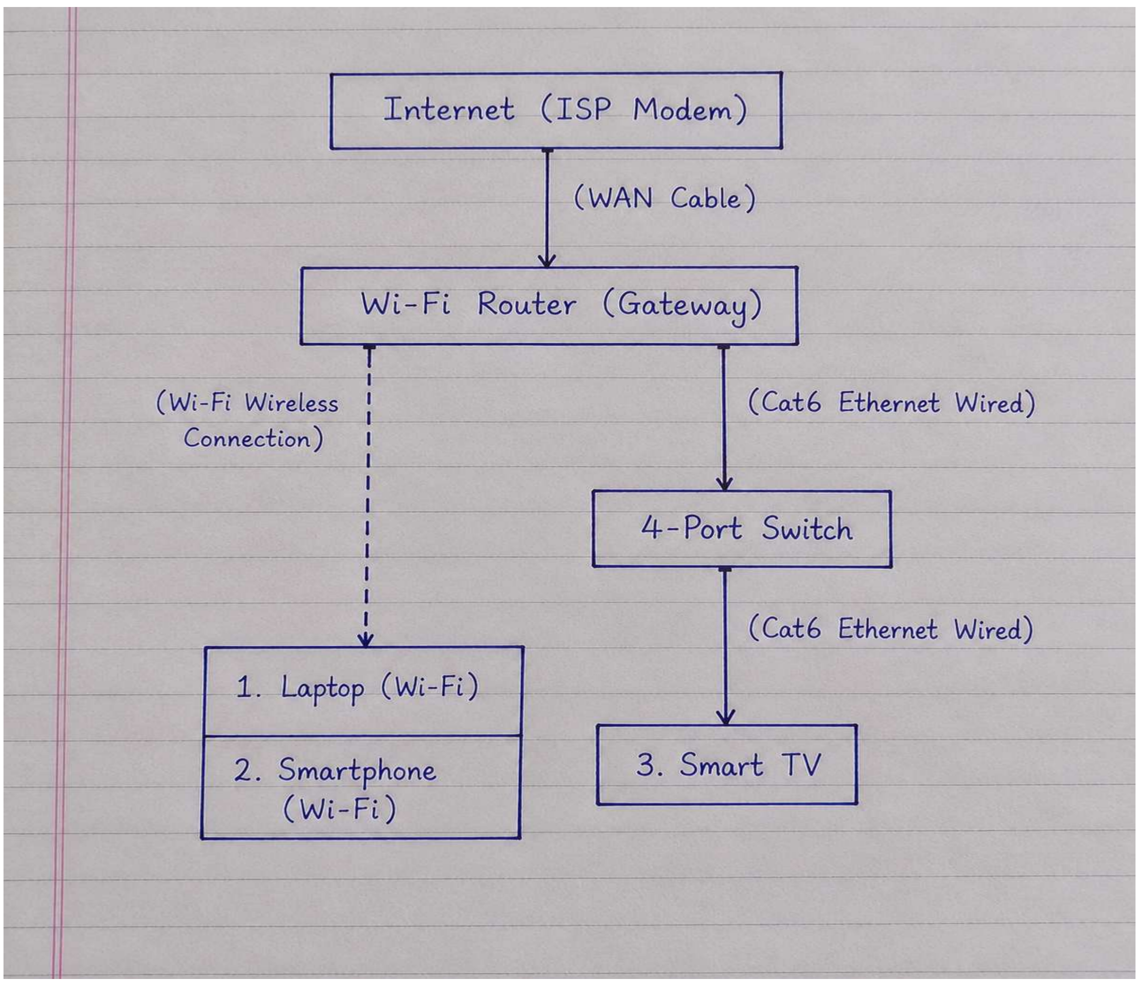
Task 1:

Home Network Topology Diagram & Connection Mapping

Objective: Diagram a standard home local area network (LAN) incorporating a router, switch, and end devices, specifying wired vs. wireless connection types..



2. Device Connection Description & Media Type:

1. **Wi-Fi Router to Internet:** Connected via ISP Fiber/Coaxial Modem using a Cat6 WAN Cable.

2. **Wi-Fi Router to Network Switch:** Connected via standard **Cat6 RJ-45 Ethernet Cable (Wired)** to expand available LAN ports.

3. **Smart TV:** Connected to the **Network Switch via Cat6 Ethernet Cable (Wired)** for ultra-stable 4K video streaming.

4. **Laptop:** Connected to the **Router via Wireless Wi-Fi (802.11ax/ac)** for mobility and high-speed internet.

5. **Smartphone:** Connected to the **Router via Wireless Wi-Fi (2.4 GHz / 5 GHz Band)** for mobile internet access.

Task 2:

Comparison of Network Devices & Daily Real-World Use Cases

Objective: Compare four foundational network devices (Router, Switch, Hub, Access Point) based on their OSI operating layer, main function, and daily application scenarios.

Comparison & Use Case Table:

Network Device	Operating OSI Layer	Key Function / Characteristics	Daily Real-World Application Use Case
Router	Layer 3 (Network Layer)	Connects distinct IP networks (LAN to WAN) and routes data packets using IP addresses.	Zomato Food Order / Google Search: Directs local app requests from home phones across the ISP network to external cloud

Network Device	Operating OSI Layer	Key Function / Characteristics	Daily Real-World Application Use Case
			application servers.
Switch	Layer 2 (Data Link Layer)	Connects multiple wired LAN devices and forwards frames directly to target ports using MAC addresses.	College Computer Lab / Multiplayer LAN Gaming: Enables ultra-low latency, high-bandwidth wired packet transfers between PCs for gaming or local file sharing.
Hub	Layer 1 (Physical Layer)	Legacy multiport repeater that broadcasts incoming data to all connected	Old Office Billing / Legacy Print Station: Connects older receipts printers or basic network monitoring

Network Device	Operating OSI Layer	Key Function / Characteristics	Daily Real-World Application Use Case
		ports indiscriminately.	devices in simple legacy setups.
Access Point (AP)	Layer 2 / Layer 1	Converts wired Ethernet signals into wireless Wi-Fi radio frequencies (802.11 standards).	YouTube 4K Streaming / Instagram Reels: Provides wireless high-speed Wi-Fi connectivity to smartphones and laptops across home and college campuses.

Task 3:

Real-World Scenario: Zomato Order Data Journey

Objective: Explain how digital data travels from a smartphone to the public internet during an online food order transaction, detailing the specific operational roles of an Access Point, Switch, and Router.

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1. Scenario: Placing an Order on Zomato App

When a user taps "Place Order" on the Zomato app while connected to a Wi-Fi network, the order payload (items, payment details, delivery address) travels through the following network device hierarchy:

1. **Smartphone to Wireless Access Point (AP):**

The smartphone converts the digital request into **Radio Frequency (RF) signals** and transmits them over the air. The **Access Point** receives these Wi-Fi signals, converts them back into electrical digital frames, and feeds them into the wired LAN.

2. **Access Point to Network Switch:**

The Access Point is connected via a Cat6 Ethernet cable to a **Layer 2 Network Switch**. The switch inspects the destination **MAC address** in the data frame and efficiently forwards the packet directly to the connected router's LAN port without broadcasting traffic to other local devices.

3. **Network Switch to Router (WAN / Gateway):**

The **Router** receives the data frame, strips the Layer 2 header, and examines the Layer 3 **IP address**. It uses **Network Address Translation (NAT)** to convert the local private IP of the smartphone into a public IP address. The router then determines the optimal path across the ISP network to route the packet to **Zomato's Cloud Data Center Servers**.

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2. Summary of Device Roles:

- **Access Point (AP):** Acts as the Wireless Bridge (converts Wi-Fi RF signals to wired Ethernet frames).
- **Switch:** Acts as the Local Traffic Traffic Cop (forwards data frames locally using MAC addresses).
- **Router:** Acts as the Gateway to the Internet (routes data packets usin

Task 4:

Remote Gaming PC Connectivity Solution Scenario

Objective: Propose two alternative network device solutions to connect an Ethernet-only gaming PC located far from the main Wi-Fi router.

Solution 1: Powerline Ethernet Adapter Kit

- **Device Suggested: Powerline Ethernet Adapter Set (e.g., AV1000/AV2000)**
- **How it Works in this Setup:**
    1. Plug **Adapter A** into a wall electrical outlet near the Wi-Fi router and connect it to one of the router's LAN ports using a short Cat6 Ethernet cable.
    2. Plug **Adapter B** into a wall electrical outlet in the gaming room near the PC.
    3. Connect a Cat6 Ethernet cable from **Adapter B** directly into the gaming PC's Ethernet port.

- **Why it's suitable:** It utilizes the existing internal electrical copper wiring of the home as a high-speed physical network cable, delivering a stable, low-latency wired connection ideal for gaming without running long Ethernet cables across rooms.

Solution 2: Wi-Fi Range Extender with Ethernet Bridge Port

- **Device Suggested: Wi-Fi Range Extender / Wireless Repeater (with Ethernet RJ-45 Port)**
- **How it Works in this Setup:**
  1. Plug the Wi-Fi extender into a wall outlet near the gaming room where it still gets a strong wireless signal from the main Wi-Fi router.
  2. Pair the extender wirelessly with the main Wi-Fi router via WPS/app setup.
  3. Connect an Ethernet cable from the **Ethernet output port on the Wi-Fi Extender** directly into the gaming PC's LAN port.
- **Why it's suitable:** The extender acts as a wireless bridge, converting the over-the-air Wi-Fi signal back into a physical wired Ethernet output for the PC.